Family Sync — Setup & Operations

Turning on optional phone-to-phone sync for the Points, Game Time, and Weekly Allowance apps — one Firebase project, one config file, one set of rules.

By Greg Snyder, Ph.D. · Perspective Behavioral Health · technical guide

One-time setup, about 20–30 minutes. No coding — you paste values into two files. Until this is done, the apps work exactly as before (each phone on its own), and the Sync tab tells parents sync isn't set up yet. This guide replaces the old "Allowance App — Sync & Hosting Setup" sheet; that design (one code hardcoded in the file, Firebase loaded for everyone) is retired.

> **The model in one breath:** GitHub/Vercel serve the app files (as today). Firestore holds a family's shared copy — but only for families who opt in. Every phone keeps its own local copy as the working and offline copy; sync merges changes between phones through Firestore. Families who never open the Sync tab download nothing extra and send nothing anywhere.

1 Decide who can start a family

This is a policy decision, not a technical one — make it before you publish rules.

Open AS SHIPPED

- Any family can tap **Start a family** and make their own code.
- Zero work for you per family.
- You (via your Firebase project) hold the synced data of whatever families opt in — strangers included, since the hub is public.

Invite-only ONE SWITCH

- No Start button. Only codes you create in the Firebase console work.
- You decide exactly which families sync; you can revoke a code anytime.
- Nothing gets stored for families you didn't invite.

> **Custody, stated plainly.** In either mode, a synced family's data (children's point/time/allowance records and behavior labels) sits in a database you control. That is ordinary personal data about minors, not PHI — *unless* PBH clinicians direct clients to use it as part of treatment, at which point it becomes PHI and the free Spark tier (no BAA) is no longer appropriate. Open mode on a public, PBH-branded site makes you custodian for anyone who opts in; invite-only keeps that population to families you've chosen. If you stay open, post a short privacy notice on the hub describing what's stored and how to delete it. (Not legal advice — worth a quick review by your counsel.)

2 Create the Firebase project

1. Go to console.firebase.google.com → **Create a project**. Name it, e.g. `pbh-family-tools`. Turn Google Analytics **off** (not needed).
2. Stay on the free **Spark** plan. It needs no credit card, which also means it can never bill you — if a quota is ever hit, sync simply pauses until the next day while every phone keeps working locally.

3 Create the database

1. Left menu → **Build → Firestore Database → Create database**.
2. Choose **Standard edition** if asked, a **US location** (e.g. `nam5` or `us-central1`), and **Start in production mode**.

4 Publish the security rules

1. Firestore Database → **Rules** tab. Select everything in the editor and delete it.
2. Open `firestore.rules` (in this package), copy all of it, paste, click **Publish**.
3. For invite-only, follow the comment inside the file: delete the MODE 1 block and un-comment MODE 2 before publishing.

What the rules enforce: a phone can open only the exact family record whose code it holds; nobody can list or browse families; nothing can be deleted from an app; and writes must have exactly the apps' own shape. If the console shows a red error on Publish, screenshot it and send it to me.

5 Paste your config

1. Firebase console → gear icon → **Project settings** → **Your apps** → web icon `</>`. Nickname it, register (skip Firebase Hosting — you use Vercel).
2. Copy the six values from the `firebaseConfig` it shows into `pbh-sync-config.js`, replacing each `PASTE_…` placeholder:

```
window.PBH_FIREBASE_CONFIG = {
  apiKey: "AIza…your value…",
  authDomain: "pbh-family-tools.firebaseapp.com",
  projectId: "pbh-family-tools",
  storageBucket: "pbh-family-tools.firebasestorage.app",
  messagingSenderId: "123456789012",
  appId: "1:123456789012:web:abc123…"
};
window.PBH_SYNC_MODE = "open";   // or "invite" — must match the rules you published
```

These values aren't secret — every Firebase web app sends them to the browser. The rules are the lock.

6 Upload to GitHub

In `greg477/pbh-family-tools` → **Add file → Upload files**, drag in and commit:

- `points-tracker.html`, `screen-time-tracker.html`, `teen-allowance-tracker.html` (replace the existing ones — same names)
- `pbh-sync-config.js` (new, at the repo root next to the apps)
- `index.html` and the three family-guide PDFs (updated text about Family Sync)

Then delete `allowance-setup-firestore-github.pdf` from the repo (open it on GitHub → ⋯ → Delete file). Keep `firestore.rules` and this guide off the public site — they're for you. Vercel redeploys automatically in about a minute.

7 Two-minute smoke test

1. **Phone 1**: open Game Time → ⚙️ → **Sync** → **Start a family**. The header pill should read **Synced**.
2. Tap **Invite parent** and text the link to **phone 2**; open it there and tap **Join family**.
3. Tap an earn item on phone 1 — it should appear on phone 2 within a second or two. Tap on both at once — both should count.
4. Put phone 2 in airplane mode, tap twice, turn airplane mode off — the pill goes **Offline · will sync** → **Synced**, and both phones agree.
5. In the Firebase console you'll see `families → (code) → apps → gametime`.

> **What was already verified before handoff:** against a stand-in database with the same transaction behavior, a 51-check suite (join/leave, invite links, conflict cases, offline, guardrails, teen screen, misconfiguration, invite-only) and a 3-phone randomized stress test (240 taps with random lag and dropouts — exact totals, no lost or doubled log entries). The smoke test above confirms the real Firebase project behaves the same way.

8 Invite-only: issuing and revoking codes

1. Firestore Database → **Data** → **Start collection** (first time) → Collection ID `allowedFamilies`.
2. **Document ID** = the code: 8–24 capital letters and numbers, no dashes — e.g. `RIVERAFAM2026`. Add any field for your own reference (e.g. `note: "Rivera family"`).
3. Give the family the code. They type it under Sync → **Join with a family code** (dashes, spaces, and lowercase are fine — the app cleans it up). The first phone uploads what it already has.
4. **Revoke** by deleting that document. Their phones show "Sync problem" and simply keep working on their own.

9 Recommended hardening (5 minutes)

Limit where your config works: Google Cloud Console → APIs & Services → Credentials → the "Browser key" Firebase created → **Application restrictions: Websites** → add your Vercel URL (e.g. `https://pbh-family-tools.vercel.app/*`) and any custom domain. Copies of your config on other sites then stop working.

10 How it works (for when someone asks)

Local first

Every tap saves to the phone's own storage under the same key as before; that copy is what the screen shows and what keeps working offline. Joining adds a second, separate key holding the family code and sync bookkeeping, so leaving sync never touches the app's data.

Conflict-safe writes

Each phone remembers the last version all phones agreed on. When it syncs, it works out what *it* changed since then and what the *family copy* changed, and combines them inside a Firestore transaction — if another phone wrote in the meantime, Firestore re-runs the merge with the newer copy. Tallies (points, banked time, balance, sticks, today's fines) add their changes, so two +1 taps make +2; settings (prices, caps, names) take the edit made, and if both phones changed the same setting differently, the most recent sync wins; activity logs merge entry by entry. Each write carries a revision number, so a phone never applies the same update twice.

Guardrails after every merge

Points and banked time never go negative; Game Time never exceeds its daily cap; Allowance never fines past the daily cap (the overflow is refunded, so that fine is stick-only — exactly what would have happened one tap after the other) and bonuses never push past the weekly amount when they're recovery-only.

Teen view-only screen

Allowance → ⚙️ → Sharing → **Send teen link**. That device joins as view-only: it listens but never writes, hides the settings gear and fine/bonus buttons, and stays a view-only screen even if opened later without the link.

11 Troubleshooting

WHAT A PARENT SEES	CAUSE	FIX
"Family Sync isn't set up on this website yet"	`pbh-sync-config.js` missing from the repo, or still has `PASTE_` values	Step 5–6
"That family code was refused" / "isn't active"	Rules not published; mode in config doesn't match the rules; invite-only code not created; typo	Step 4, 5, 8
Pill says "Offline · will sync"	No connection	Nothing — it catches up on its own
iPhone: joined in Safari, but the home-screen icon isn't synced	iOS keeps home-screen apps' storage separate from Safari	Join from inside the home-screen app — or open the invite link, then Add to Home Screen from that page
A parent's phone accidentally became a teen view	It opened the teen link	Clear that site's data (iPhone: Settings → Apps → Safari → Advanced → Website Data), then rejoin as a parent
Family wants to start over	—	Each phone: Sync → Stop syncing. Optionally delete `families → (code)` in the console.

12 Data map

WHERE	WHAT
Firestore `families/{CODE}/apps/{points | gametime | allowance}`	`state` (the app's data), `app`, `v`, `rev` (revision), `updatedAt`
Firestore `allowedFamilies/{CODE}`	Invite-only codes you issue (phones can't read this)
Phone: the app's existing key (e.g. `screen-time-app-v1`)	The app's data — unchanged format; works with or without sync
Phone: same key + `:sync`	Family code, last agreed version, revision, unsynced flag (only on phones that joined)
Phone: same key + `:before-sync`	Snapshot of that phone's data at the moment it joined — insurance



Created by **Greg Snyder, Ph.D.** — a Perspective Behavioral Health resource. © 2026 Greg Snyder, Ph.D. All rights reserved. Console wording reflects Firebase and Google Cloud as of 2026; if a button name differs, look for the nearest equivalent.